BrightLearn Data Analytics

Research Assignment 1

Foundations of Data Analytics & Data Science

Course	Data Analytics & Data Science
Assignment No.	Research Assignment 1
Target Audience	Beginner Students
Due Date	To be confirmed by instructor
Format	Handwritten answers — scan and email your completed work

Instructions

Read each question carefully and research the answer before writing.

- All answers must be handwritten — typed submissions will not be accepted.
- Write in full sentences and use your own words.
- Include diagrams or examples where relevant — this will earn you bonus marks.
- Cite your sources at the bottom of each section.
- Scan your completed assignment and submit it before the due date.

Section A: Data Fundamentals

Research Tip: *Start with official documentation from platforms like IBM, Google Cloud, AWS, or Microsoft Azure — they offer excellent beginner-friendly explanations.*

1. What is a Database? Define it in your own words and explain why databases are essential in the modern world. Provide two real-world examples of where databases are used.

IS a structured collection of data that IS stored

electronically and can be easily accessed and updated. Database is essential because every system we use depends on data.

2. What is a Data Warehouse? How does it differ from a regular database? Why would a company use a data warehouse?

Is a central place where we store multiple database. ~~thats~~ warehouse its different because it is designed to store large data where company can keep all its historical and current data so that analyst can study and understand it.

3. What is a Data Lake? Explain the concept and describe how it differs from a data warehouse. When would a business prefer a data lake over a data warehouse?

Data lake stores raw, unprocessed data in different format while data warehouse stores clean and structured data. Businesses prefers a data lake when they need to store large amount of different type of data, however for structured data repating data warehouse is suitable.

4. What is a Data Mart? How does it relate to a data warehouse and who typically uses it?

Is a smaller section of a data warehouse that focuses on a specific department or business function.

Data mart can be used by business analyst and manag for decision making.

5. Explain the difference between Structured, Semi-Structured, and Unstructured Data. Provide one example of each.

* Structured data is stored in a table with rows/columns in a consistent format e.g Bank statement.
* Semi-Structured data is not organized but have a structure e.g JSON file.

* Unstructured data - has no specific format/structure e.g videos/images.

6. What is Data Modeling? Why is it important when designing a system that works with data? Describe at least two types of data models.

Is the process of designing how data is stored, organized and connected in a system. It prevent messy or duplicate da

1. Conceptual model - high level view
2. Logical model → It is more detailed

7. What is Data Mining? How is it used in business? Provide two real-world examples of data mining in action.

Data mining is the process of analyzing large set of data to discover patterns, trends and useful information.

* To understand customer behavior and preferences.
* Predict trends and future sales.

8. What is a Database Management System (DBMS)? Name and briefly describe four popular DBMS platforms (e.g., MySQL, PostgreSQL, Oracle, MongoDB) and explain a suitable use case for each.

A DBMS is software used to store, manage and control data in a database. Popular DBMS include MySQL for web applicatio PostgreSQL for secure and complex system, oracle database for large enterprise systems and MongoDB is flexible and unstructured data application.

9. What is ETL (Extract, Transform, Load)? Explain each step and describe why this process is important in data analytics.

ETL is data process used to collect data from different source, clean it, and store it in a system where it can be easily analyzed. It is important because it improve data quality ensure consistency and helps with decision making

10. What are common data storage formats used in analytics (e.g., CSV, JSON, Parquet, Avro)? Briefly describe each and state when you would use one over another.

* CSV is a simple file format where data is stored in rows and columns separated by commas.

* JSON store data in a key-value making it flexible and readable for machines and humans. (web applicata

* Parquet is a column-based storage format designe for fast processing of large datasets.

* Avro is a row-based format that include a schema for data structure and supports data.

Section B: Artificial Intelligence & Machine Learning

Research Tip: *Explore resources like Google's Machine Learning Crash Course, IBM's AI explanations, or OpenAI's blog to build a strong foundational understanding.*

11. What is Artificial Intelligence (AI)? Define it and explain the difference between Narrow AI (Weak AI) and General AI (Strong AI). Provide one example of each.

AI refers to machine or system that can perform tasks that requires human brains such as learning and reasoni

* Narrow AI - can only do one specific job e.g google map showing you directions
* General AI - can think like a real human ~~do~~ do homework

12. What is Machine Learning (ML)? How does it differ from traditional programming? Describe the three main types of Machine Learning: Supervised, Unsupervised, and Reinforcement Learning.

* is when a computer learns from experience data insteed of bieng told everything.
* Supervised - computer is taught using correct answers
* unsupervised - computer finds patterns on its own
* Reinforcement - Learns through trail and errors to improve its mistakes.

13. What is Deep Learning? How does it relate to Machine Learning and Artificial Intelligence? What are Neural Networks and how do they work?

* Deep learning is a powerful part of machine learning that uses neural netwaks to learn from large data and solve complex problem, and it is a key part of AI. They relate to AI, ML because machine learns from data instead of manual programing.

14. What is Natural Language Processing (NLP)? Describe what it is and provide three real-world applications of NLP that you use in everyday life.

* is when computers can understand human language

* voice assistant (google assistant)
* chatGPT (chatbots)
* Google translate.

15. What is a Large Language Model (LLM)? How is it different from a standard Machine Learning model? Name two well-known LLMs and explain what they are used for.

* LLM is a powerful type of machine learning model designed to understand and generate human language and it is more advanced and flexible than standard ML.
* ChatGPT and Google gemini (chatgpt helps with coding and gemini assist with writing content)

16. What is Generative AI? How does it work at a high level? Name three tools that use Generative AI and describe what each one does.

* is a branch of AI focused on creating new content and it works through pattern recognition
* DALL·E - create images from text
* Github - helps ~~with~~ developer with real time codes
* grammarly - helps in improving writing, grammer and tone.

17. What is a Training Dataset? Why is the quality of training data so important when building an AI model? What could go wrong if the training data is biased or incorrect?

- Training dataset is the collection of data that we use to teach an AI model how to make decision.
* The quality of training data is very impatant because the AI can only learn from what it is given. but if the data is wrong AI will also become wrong.

18. What is Computer Vision? How do machines 'see' and interpret images? Provide two real-world use cases of computer vision.

* Computer vision is a type of AI that allows machines to see and understand images or videos
* Face recognition and self driving car

Section C: APIs, Tools & Developer Concepts

Research Tip: *Search for 'What is an API for beginners' on YouTube or read documentation from platforms like Postman, RapidAPI, or Anthropic.*

19. What is an API (Application Programming Interface)? Explain what it is using a simple real-world analogy. Why are APIs important in modern software and data systems?

* API is a tool that allows two different software to talk to each other and share information.
* Me (user) are the customer
* kitchen is the system data base
* waiter (API) it takes the order then kitchen after brings it back.

20. What is an API Key? What is its purpose? How should it be stored and protected, and what could happen if it is exposed publicly?

* An API key is a unique code used to identify and authen a user when using API, also controls who can access API.
* it must be secured because it keeps secrets and private.
* if it is exposed hacker may steal data and misuse it.

21. What is a REST API? What does REST stand for and what are the main HTTP methods (GET, POST, PUT, DELETE) used in REST APIs? Briefly describe each method.

* A rest API is a way for system to communicate over the internet using simple web rules.
* Rest (representation state transfer) - Put - update data.
* Post - send and create
* delete - Remove data

22. What is JSON (JavaScript Object Notation)? Why is it commonly used when working with APIs and data exchange? Write a simple example of a JSON object.

* JSON is a simple format to store and exchange data
* it commonly use because it is easy for computer to read / write.
* works well with most programming language.


```
[
"Name", "Hitekan",
"Age"; 25,
]
```

23. What is Prompt Engineering? Why is it an important skill when working with AI tools like ChatGPT or Claude? Provide two examples of a poorly written prompt and an improved version of each.

* promat is the process of creating clear and effictive instructions for AI system. It is important because it underste exactly what you want and gives you accurate answers.
* poor promat - write about data
* improved → example the simple terms for data.

24. What is a MAP function? Explain what it does in programming and data processing. Provide a simple example in either Python or JavaScript.

* map function is used in programming to go through a list of items and apply a functions to each item also helps process all items in a list quickly.

e.g Number = [1,2,3]
results = list(map(lamba x: x *2, numbers))
print (results)

25. What is Cloud Computing? Name the three main cloud service models (IaaS, PaaS, SaaS) and briefly explain each. Give one example of a product for each model.

* Cloud computing is when you use internet-based services to stor data, run applications or access software using your own compute
* Iaas (infrastructure as a service) - Amazon web service
* Paas (platform as a service) - google app engine
* Saas (software as a service) - microsoft 365

26. What is a Version Control System? What is Git and why is it widely used by data analysts and developers? Briefly explain what a 'commit' and a 'branch' are.

* VCS is a tool that helps developers track changes in code overtime.
* Git tracks changes
* commit fixes login bug

* Branch - is a tool that can separate copy of your project without affecting the main work.

27. What is a Jupyter Notebook? What programming language is it most commonly used with? Why is it popular among Data Analysts and Data Scientists?

* Jupter is a tool used to write and run code step by step in a web brawser, it is most commonly used with pbython.
* it allows you to see results quickly and analysis data.
* Helps with testing and experimenting step by step.

28. What is Python? Why is it considered one of the most popular programming languages for Data Analytics and AI? Name four Python libraries commonly used in data work and briefly describe each.

* python is a high-level interpreted and general pulpose programming language. used to write instruction for computer

* It is popular because it is easy to read and learn
* It is widely use in AI and data science
* it saves time because you can do complex tasks using one code.

Python Libaries

1. Numpy - use for working with numbers and calculations
2. Pandas - organize and analys data
3. Scikit-learn - used for machine learning
4. matplotlib - used for creating graphs and charts.

Section D: Data Ethics, Governance & Careers

Research Tip: *Explore the World Economic Forum, GDPR documentation, and LinkedIn Learning for content on data ethics and career pathways in data.*

29. What is Data Privacy? Why is it important? Research the GDPR (General Data Protection Regulation) and explain what it is and which countries it applies to.

* Data privacy is the protection of personal information so that it is not misused, shared without permission.
* GDPR is a law created to protect people's personal data and privacy online, and it applies in all countries in European union.

30. What is Data Ethics? Provide three examples of unethical use of data or AI, and explain what makes each example problematic.

* Data Ethic is using data and AI in a fair, responsible way.
* Sharing personal information without permission
* AI making unfair decision by rejecting job application based on race or gender.

31. What is Data Governance? Define it and explain the key components of a good data governance framework.

* Data governance is a system of rules, process and responsibility that control how data is managed in an organisation.
* Data quality – make sure data is correct and reliable.
* Data security – Protect data from unauthorized access/theft

32. What is Bias in AI? Explain what it means for an AI model to be biased. Provide a real-world example of AI bias and describe the potential consequences.

* Bias AI is a system that produce unfair or unbalanced results because it was programmed to do so.
* it makes unfair decision also favour one group over another.

e.g AI hiring system

* unfair treatment of people
* Damage of company policies

33. What are the key differences between a Data Analyst, a Data Scientist, a Data Engineer, and an AI Engineer? Create a simple table comparing their roles, skills, and tools.

Roles	Role	Skills	Tools
Data analyst	explain what happened in the past based on data	Data cleaning, charts, repats and statistics	Excel, SQL, PowerBI Tablue, Python
Data Scientist	study data and predict Future outcomes	machine learning, Programing, data modeling	Python, SQL, Scikit Databricks, Github.
Data Engineer	Builds data system	ETL Pipelines, Cloud computing	SQL, Apache, Python AWS
AI Engineer	Builds intelligence systems	machine Learning Deep learning Software Engineering	Python, Tensorflow Pytorch

34. What is Business Intelligence (BI)? Name two popular BI tools (e.g., Power BI, Tableau) and explain how they help businesses make data-driven decisions.

* BI is the process of collecting, analyzing and presenting data to help business make better decisions.
* microsoft Power BI - Create dashboards and reports
* Tableau - Turn raw data into visual dashboards.

Section E: Personal Reflection

Answer the following reflection questions thoughtfully and honestly. There is no right or wrong answer — we want to know what you think!

35. Which concept from this assignment did you find most interesting or surprising? Explain why.

It was Python and AI, because Python is ~~more~~ simple and easy to learn even beginners can learn it, while AI is surprising because it allows computers to learn and make human decisions.

36. Which concept did you find most difficult to understand? What resources did you use to help you understand it better?

most difficult concerts. are [section B, C, D], I used chatbot and introduction of SGL book from the portal.

37. How do you see Data and AI affecting your future career or everyday life? Be specific with examples.

Data and AI already have a good impact on both my future career and everyday life because I use data everyday and AI its helping me at work to be more efficient and make better decisions. I also learn how to write, code and grammer.

References

List all the sources you used to complete this assignment. Use the format below:

Author/Organisation. (Year). Title of article or webpage. Website Name. URL

1. OpenAI (2024) chatGPT documents
2. google cloud
3. BrightLearn virtual class notes.
4.
5.
6.
7.
8.

Good luck! We believe in you.
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